DATA STORYTELLING REPORT

ApexPlanet Data Analytics Internship — Task 4

1. Executive Summary

This report converts the supplied sales dataset into a business narrative. The dataset contains 1,000 records and 12 columns, with total recorded sales of ■139,399,439.65 and 5,435 units sold. Electronics is the highest-revenue category (■50,778,581.70), and Patna is the highest-revenue city (■19,285,966.89). The one-way ANOVA test returned F = 0.7171 and p = 0.580286; at α = 0.05, the conclusion is to fail to reject the null hypothesis.

2. Business Problem and Objectives

The business needs a clear view of where sales are concentrated, how performance changes over time, and whether average transaction sales differ across product categories. The objectives are to summarize sales, identify high-performing categories and cities, review monthly patterns, and apply a basic statistical test to validate one observed difference.

3. Dataset and Method

The workbook contains 1,000 rows and 12 columns. The analysis uses Total_Sales as the primary sales measure, Category for product grouping, City for geographic grouping, and Order_Date for monthly aggregation. The date range is 01 Jan 2025 to 01 Jan 2026. Group totals and averages are descriptive summaries. One-way ANOVA compares mean transaction sales across categories at a 5% significance level.

4. Key Findings

Measure	Finding
Total sales	■139,399,439.65
Average transaction sales	■139,399.44
Units sold	5,435
Top category by revenue	Electronics — ■50,778,581.70 (36.4%)
Top city by revenue	Patna — ■19,285,966.89 (13.8%)
Strongest month	2025-03 — ■13,059,899.94
ANOVA p-value	0.580286

5. Business Interpretation

Electronics leads in total revenue, making it a useful category to investigate for inventory, marketing, and customer-demand planning. Patna leads in city-level revenue and may be a useful location for deeper customer and product-mix analysis. The strongest month is 2025-03; the business should examine campaigns, seasonal demand, and unusual large orders before treating this peak as a recurring pattern.

6. Statistical Validation

H■: Mean transaction sales are equal across all categories.
H■: At least one category mean differs.
Test: One-way ANOVA; α = 0.05; F = 0.7171; p = 0.580286.
Decision: Fail to reject H■. There is insufficient evidence at the 5% level that category mean transaction sales

differ.

7. Recommendations

1. Review revenue alongside transaction count, average order value, units sold, and profit margin.
2. Investigate the top category and city to understand customer segments and product mix.
3. Monitor monthly sales in a dashboard and compare changes with campaigns and seasonality.
4. Document how missing values and duplicate order identifiers are handled before reporting results.
5. Check ANOVA assumptions; if results are significant, conduct an appropriate post-hoc test. For causal claims, use a controlled experiment.

8. Limitations

This is an observational dataset. It can describe patterns but cannot establish that a category or city caused higher sales. ANOVA relies on independence and other distributional assumptions. The analysis does not include profit margin, marketing cost, or customer acquisition cost, so revenue alone should not be treated as profitability.

Source: user-provided ApexPlanet sales dataset. Figures are calculated from the workbook.